



C Programming

CERTIFICATION COURSE

**Master C Programming Skills and
Elevate Your Career to New Level !**

C Programming

The C Programming Course, offered by W3 Technologies, will equip you with the skills to write efficient code using procedural programming techniques. You'll learn to break down complex problems into manageable components, focusing on C syntax and best practices for clear and maintainable programs. This course emphasizes memory management, pointers, and modular design to build a solid foundation in C programming.

Why W3 Technologies

Training at W3 Technologies equips you to excel in today's competitive world. From the very beginning of your journey with us, you'll experience continuous growth through learning. We provide support throughout the course, while encouraging you to become independent and confident in your skills.



Lower Learning Cost

Learning with us is budget-friendly. Our courses are affordable yet provide comprehensive and engaging content.



Expert Instructors

Learning with us means benefiting from top-tier expertise. Our expert instructors provide deep knowledge and engaging lessons.



Certification

Earn an industry-recognized certification upon completing the course, enhancing your resume.

Introduction

- Why Programming
- History of C Language
- Hello World Program

Variables and Data Types

- Identifiers in C
- Variables and Data Types
- Constants

Console IO Operations

- printf and scanf
- Unformatted IO Functions

Operators and Expressions

- Expressions and Arithmetic Operators
- Relational and Logical Operators
- Bitwise Operators

Control Flow Statements

- If Statement
- Switch Statement
- Unconditional Branching using goto statement
- While Loop
- Do While Loop
- For Loop
- Break and Continue

Working with Functions

- Introduction and Writing Functions
- Scope of Variables, Storage Classes
- Pass by Value and reference
- Recursion

Working with Arrays

- Arrays Declaration and Initialization Properties
- Sample Programs using Arrays
- Arrays as Function Parameters
- 2-Dimensional Array

Pointers

- Introduction to Pointers
- Pointers as Function Parameter
- Pointer Arithmetic
- Pointers and Arrays
- Function Pointers
- Dynamic Memory Allocation using malloc
- calloc and comparison with malloc

String Handling

- Introduction to Strings
- Sample Program
- More Sample Programs
- Standard String Library Functions
- Array of String

Structures and Unions

- Declaring and Instantiating Structures
- Structure as Parameter and Pointer to Structure
- Enumerated Data Type
- Union
- Bit Fields

File Handling

- What is a Stream
- File Handling-Writing and Reading Characters
- Writing and Reading Structure in Text Format
- Writing and Reading in Binary Format

Pre-Processor Directives

- Understanding Pre-Processor directives
- Header Files and Project

Command Line Arguments and Variable Arguments

- Command Line Argument
- Variable Number of Arguments

Searching and Sorting

- Linear Search
- Iterative Binary Search
- Recursive Binary Search
- Bubble Sort
- Selection Sort
- Insertion Sort
- Quick Sort
- Merge Sort

Linked List

- Introduction to Linked List
- Single Linked List
- Double Linked List
- Circular Linked List

Stack

- Introduction to Data Structures
- Stack Using Arrays
- Stack Using Linked List
- Infix to Postfix Expressions

Queue

- Queue Using Arrays
- Queue Using Linked List

Trees

- Introduction to Trees
- Programming Tree – 1
- Programming Tree – 2

Thank You for Choosing **W3 Technologies**

Thank you for choosing to explore this course from W3 Technologies. We hope you find it insightful and valuable as you embark on your learning journey.

At W3 Technologies, we believe that learning is a continuous process, and we are excited to be part of your journey towards gaining new skills and knowledge.



www.w3tec.com



info@w3tec.com



+91 74169 39333



**4th Floor, LR Shine Building, Vrc,
above IDBI Bank, beside 1-Town
Police Station, Somasekara
Puram, Nellore, Andhra Pradesh
524001.**

